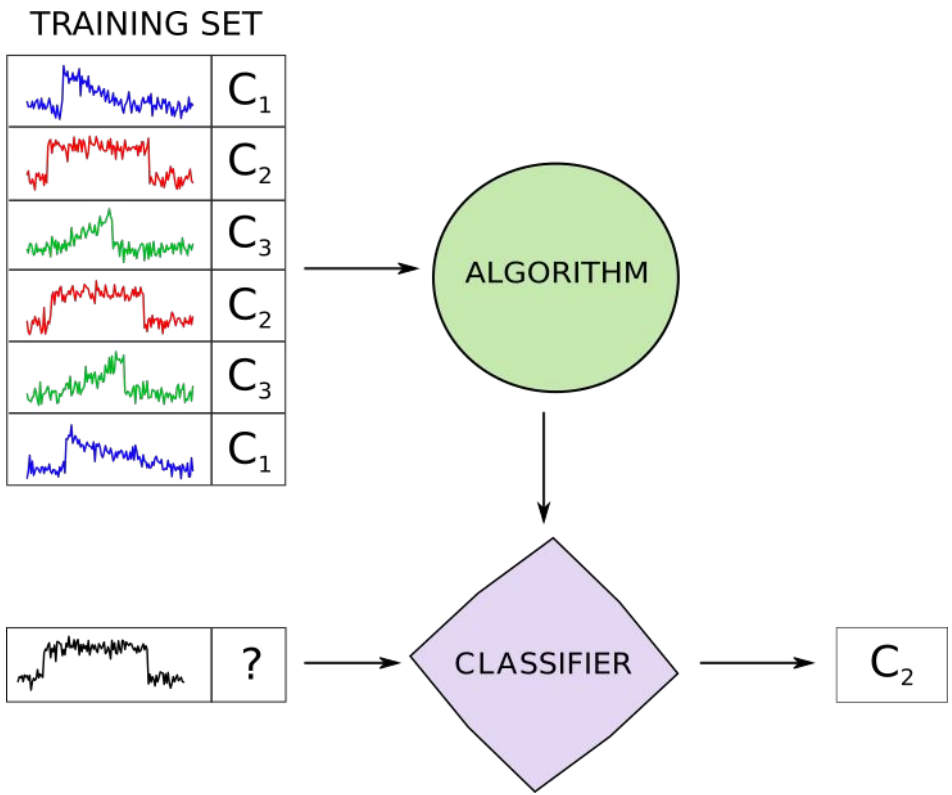
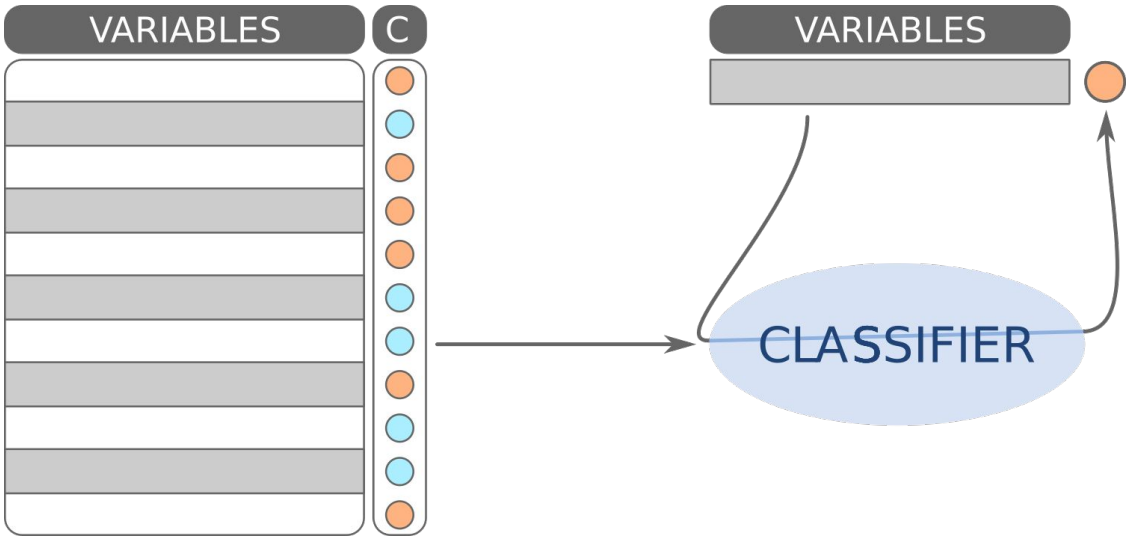
Time series classification


TRAINING SET
C1
C2
C3
C2
C3
C1
ALGORITHM
?
CLASSIFIER
C2

Differences with the classical problem

- Temporal correlation.
- High dimensionality.
- Noise.



Taxonomy I of TSC (Esling et al. 2012)

- Feature based classification.
- Model based classification.
- Distance based classification.

Taxonomy II of TSC (Bagnall et al. 2017)

Data Min Knowl Disc (2017) 31:606–660
DOI 10.1007/s10618-016-0483-9

CrossMark

The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances

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Aaron Bostrom[1] · James Large[1] · Eamonn Keogh[2]

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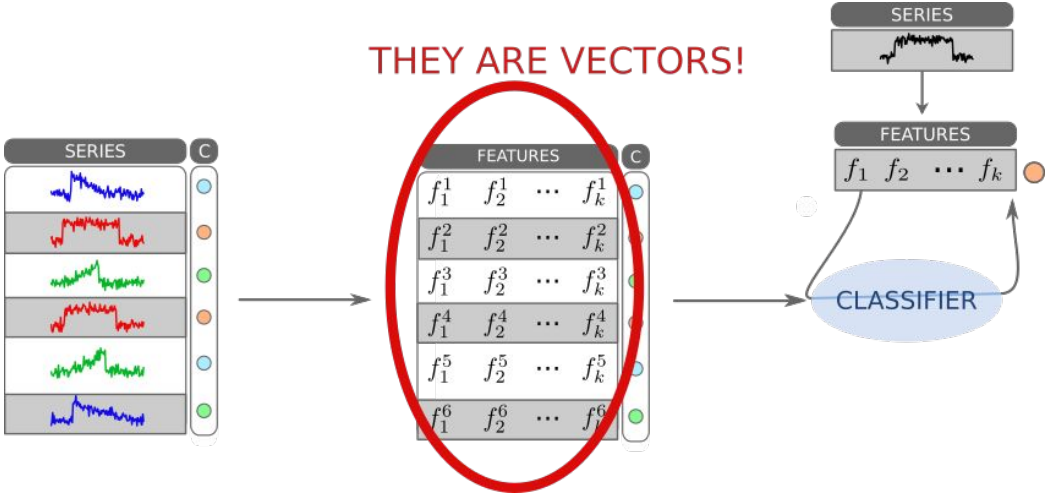
PROJECT 3

[illegible] there ha[illegible] bee[illegible] ber of ne[illegible]me series c[illegible] [illegible]orithm[illegible] prop[illegible]d in th[illegible]era[illegible]. The[illegible]gorithms [illegible]e been evaluate[illegible] on sub[illegible]s [illegible]ets [illegible] Uni[illegible] C[illegible]a, R[illegible]rside time classification archive. The archive has recently been expanded to 85 data sets, over half of which have been donated by researchers at the University of East Anglia. Aspects of previous evaluations have made comparisons between algorithms difficult. For example, several different programming languages have been used, experiments involved a single train/test split and some used normalised data whilst others did not. The relaunch of the archive provides a timely opportunity to thoroughly evaluate algorithms on a larger number of datasets. We have implemented 18 recently proposed algorithms in a common Java framework and compared them against two standard benchmark classifiers (and each other) by performing 100 resampling experiments on each of the 85 datasets. We use these results to test several hypotheses relating to whether the algorithms are significantly more accurate than the benchmarks and each other. Our results indicate that only nine of these algorithms are significantly more accurate than both benchmarks and that one classifier, the collective of transformation ensembles, is significantly more accurate than all of the others. All of our experiments and results are reproducible: we release all of our code, results and experimental details and we hope these experiments form the basis for more robust testing of new algorithms in the future.

Taxonomy of TSC (Esling et al. 2012)

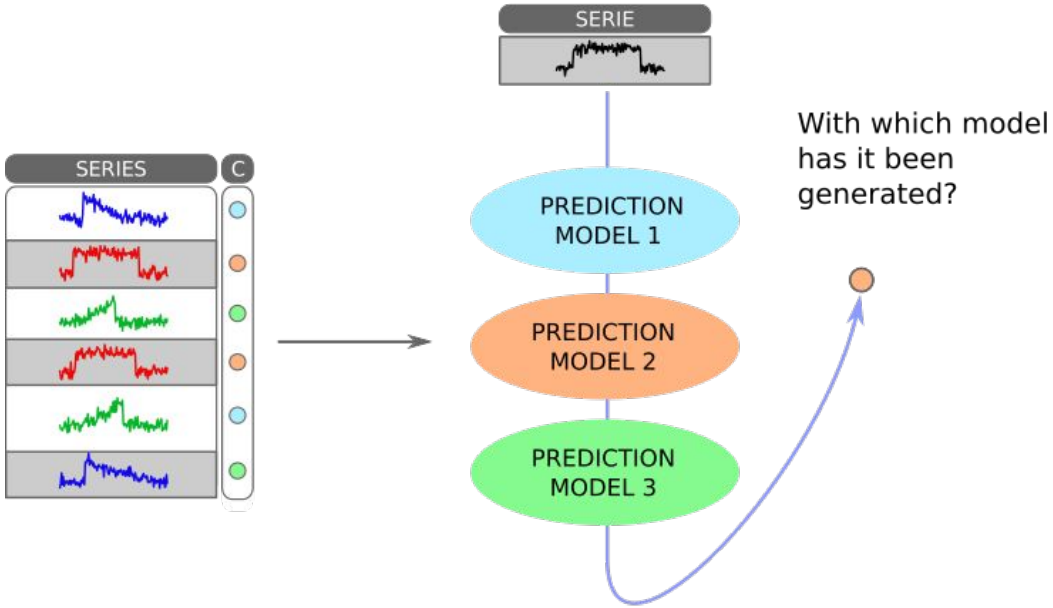
- Feature based classification.
- Model based classification.
- Distance based classification.

Feature based classification



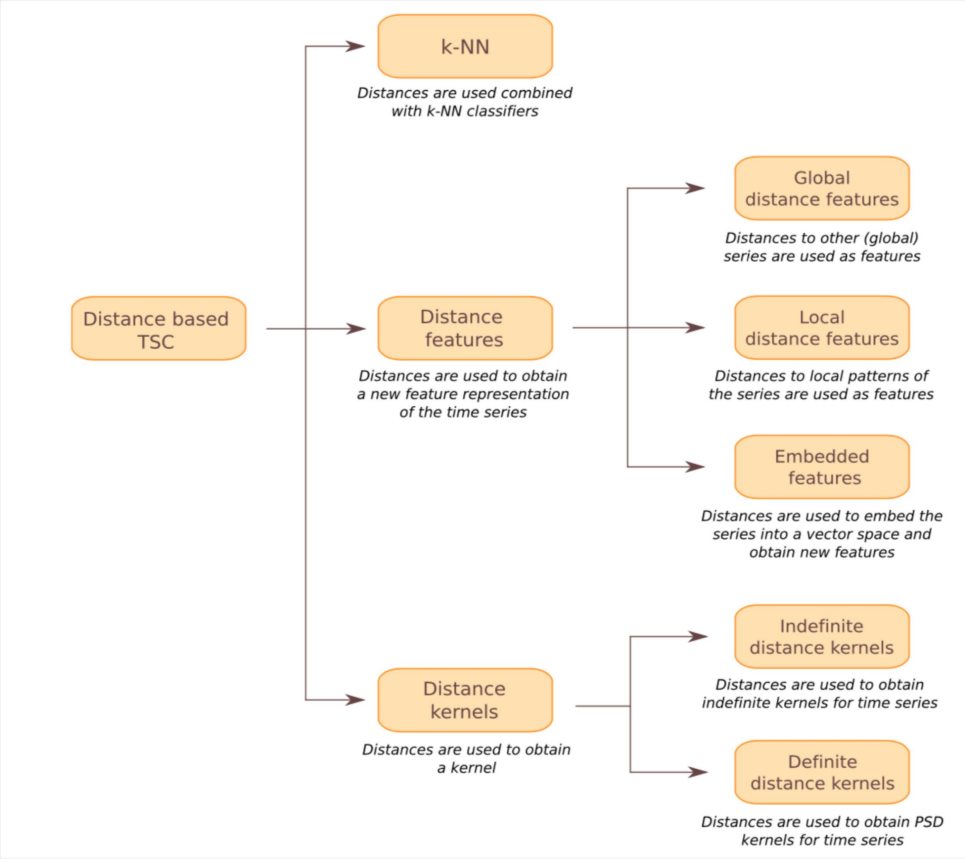
- Representation does not depend on the number of training series.
- Interpretable.
- What features?

Model based classification

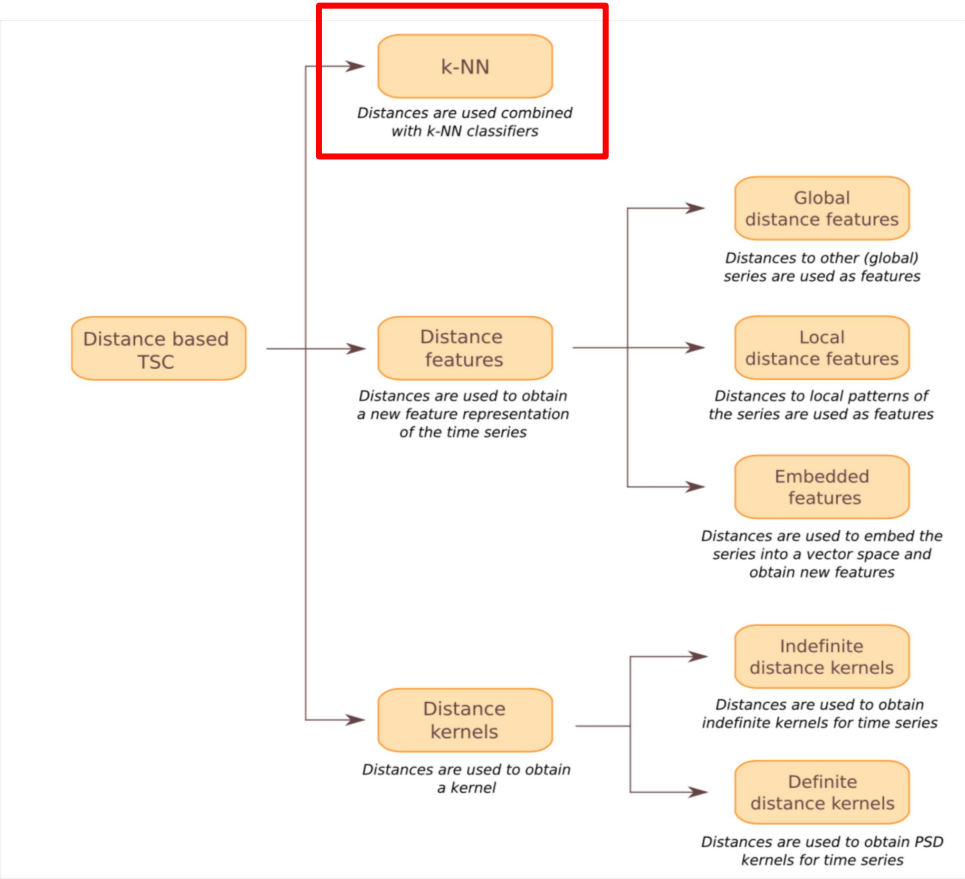


- Good results with adequate models
- What model? Does it exist?

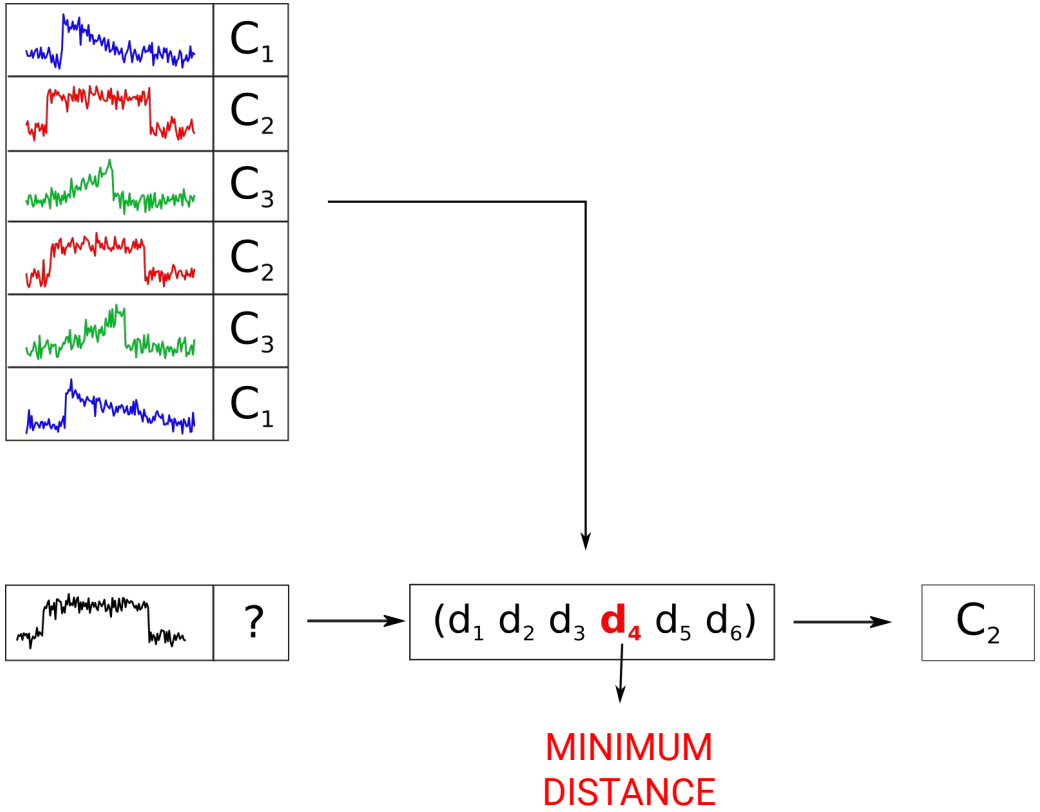
Distance based classification (Abanda el al. 2019)



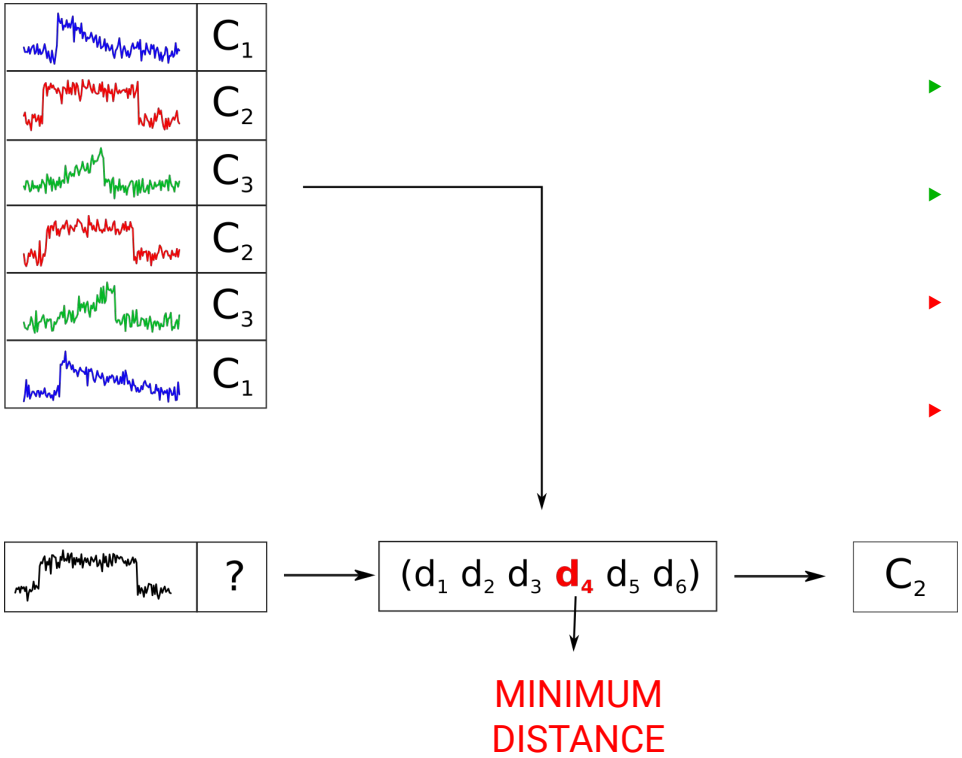
Distance based classification (Abanda el al. 2019)



Nearest neighbor algorithm

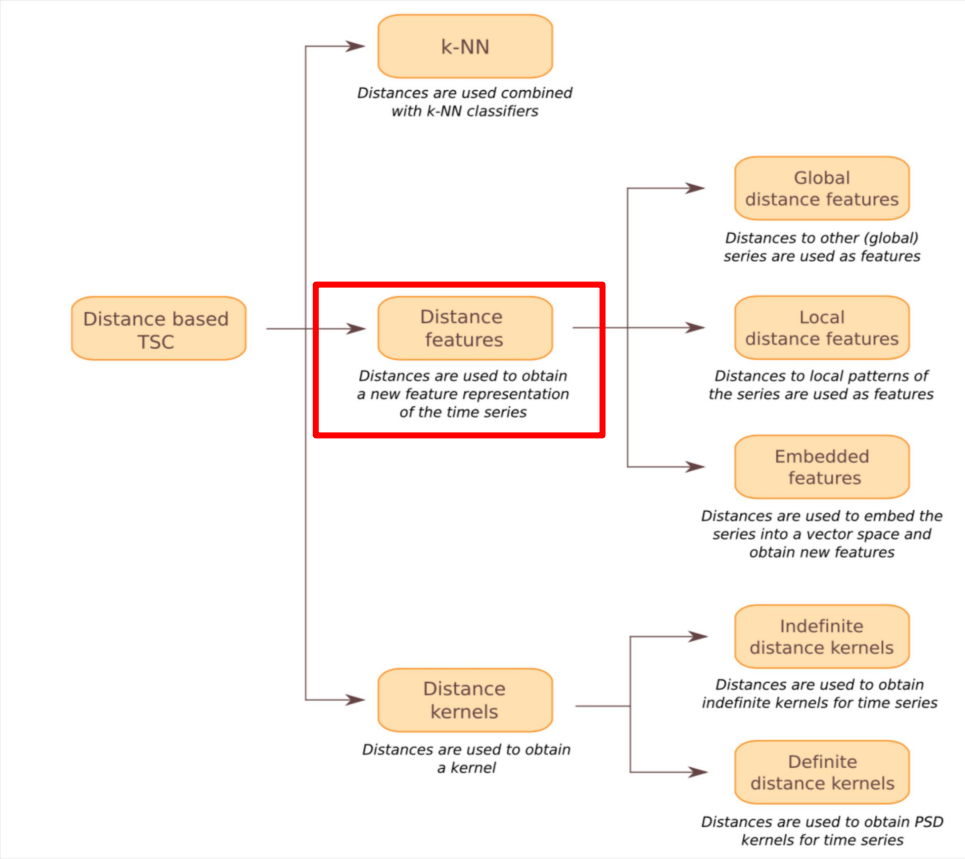


Nearest neighbor algorithm

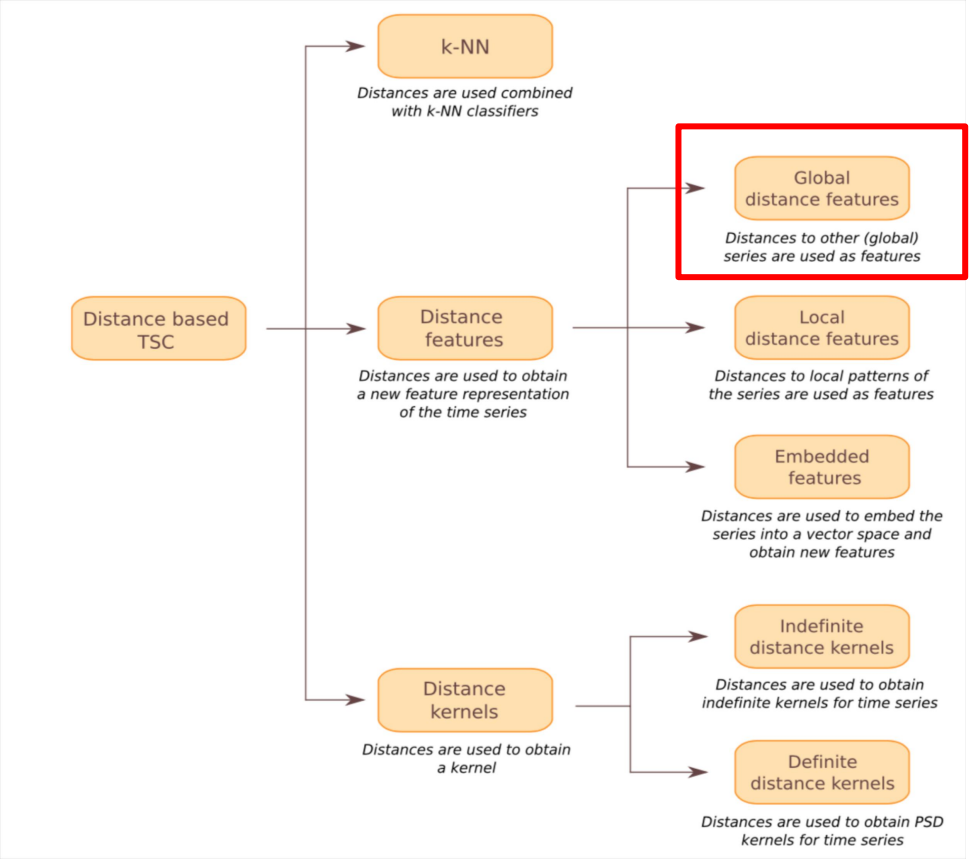


- Intuitive.
- Good results with large datasets.
- Computational cost.
- Which distance?

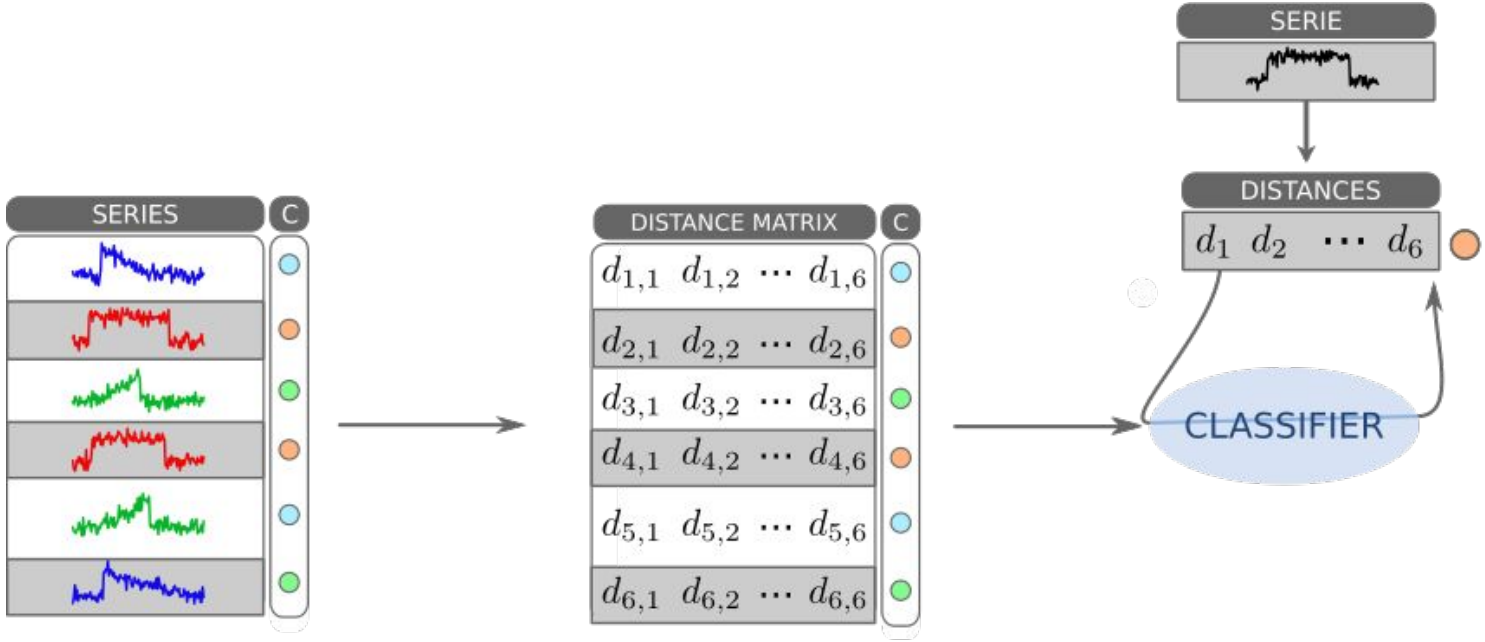
Distance based classification (Abanda el al. 2019)



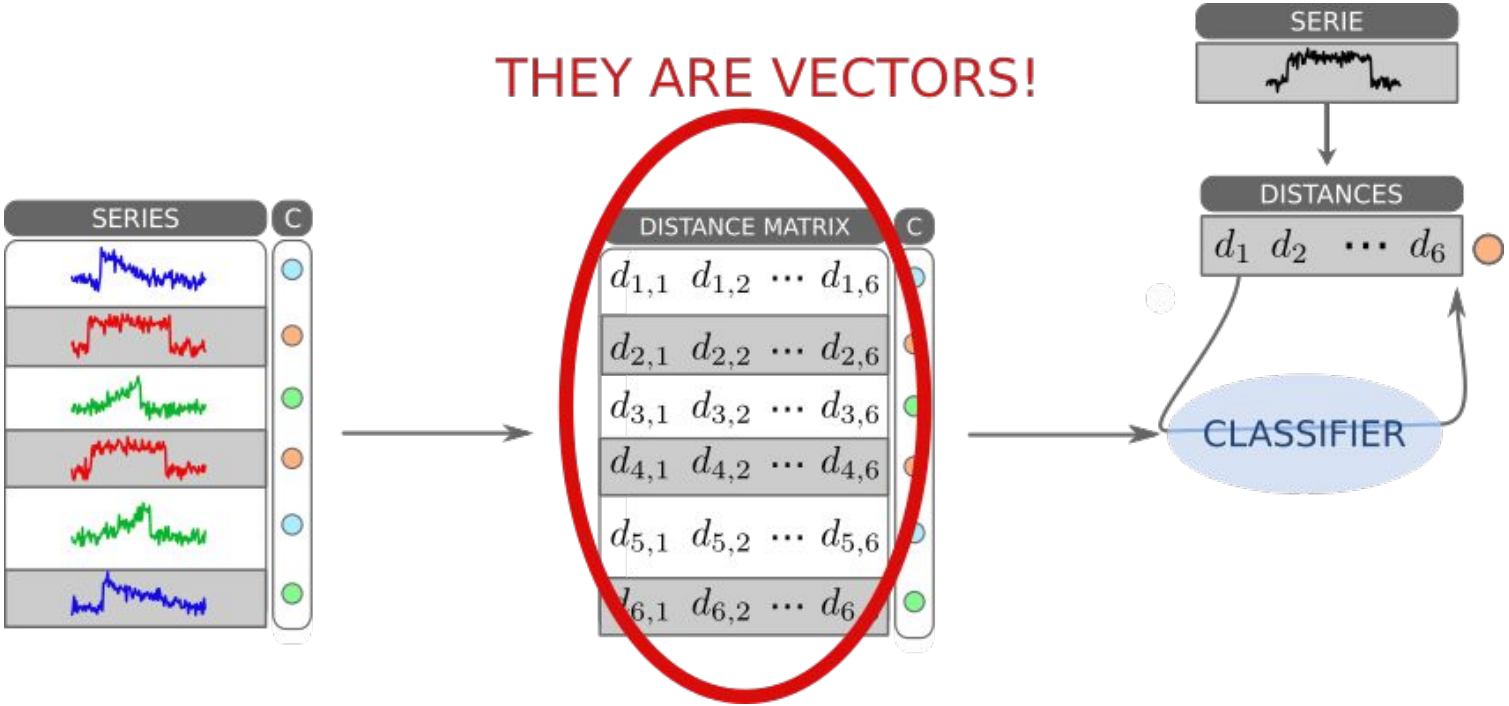
Distance based classification (Abanda el al. 2019)



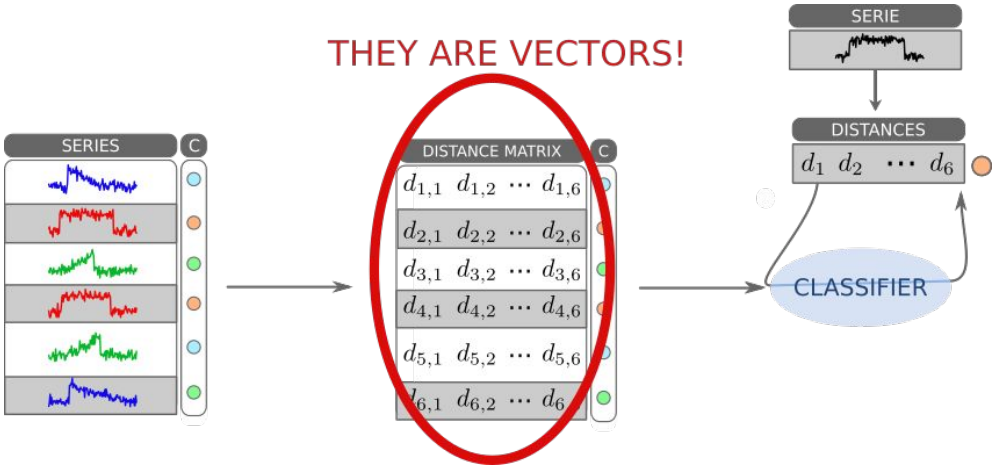
Distance based global features



Distance based global features

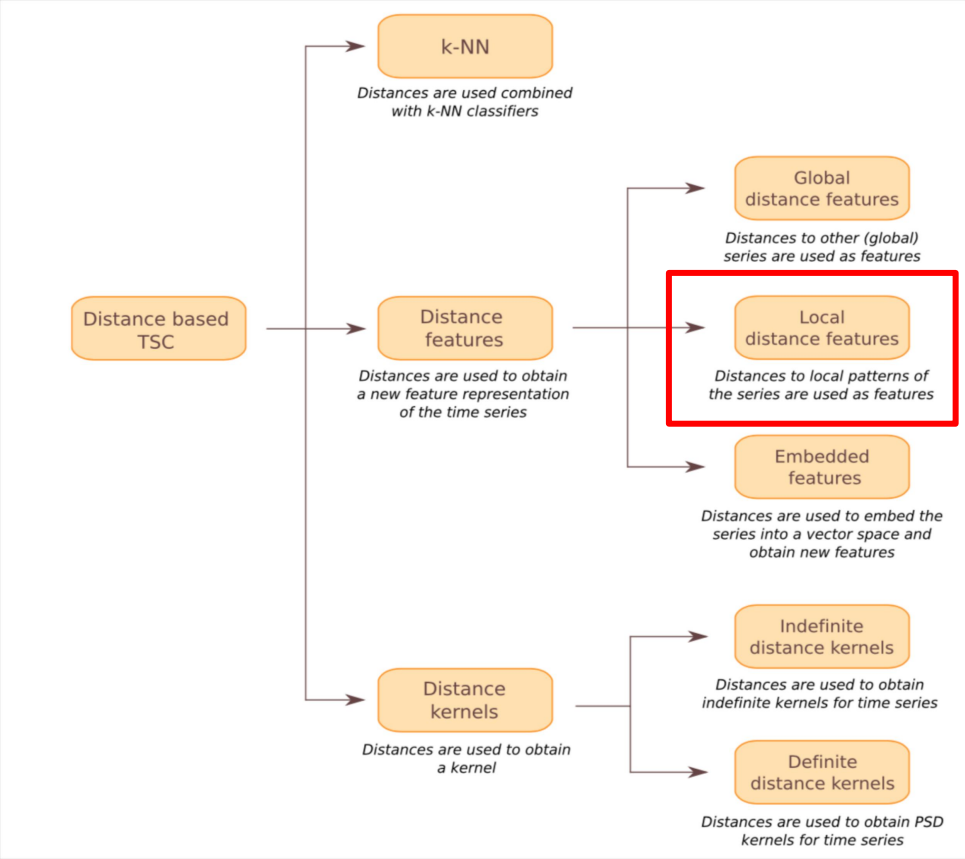


Distance based global features

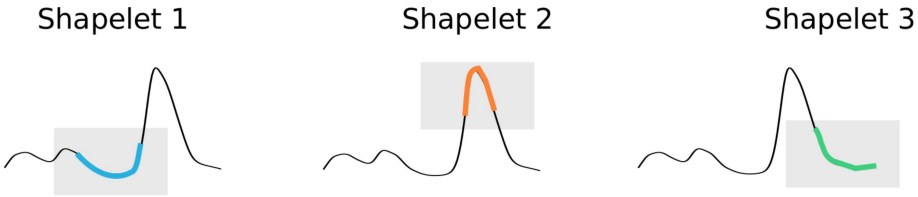


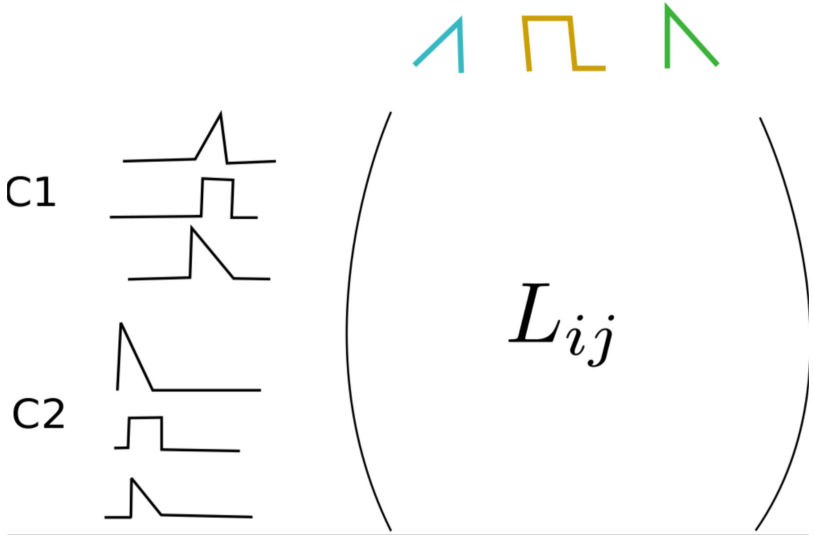
- Any standard classifier can be used.
- Representation depends on the size of the training set.
- High computational cost.

Distance based classification (Abanda el al. 2019)

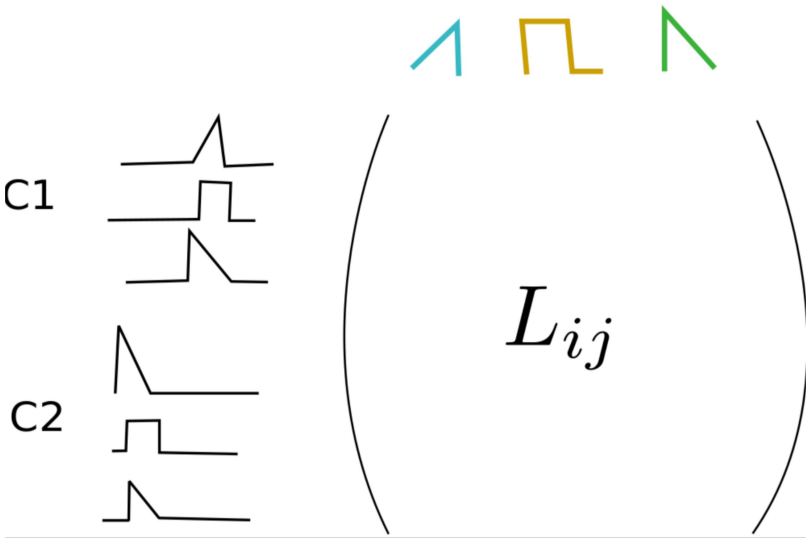


Distance based local features



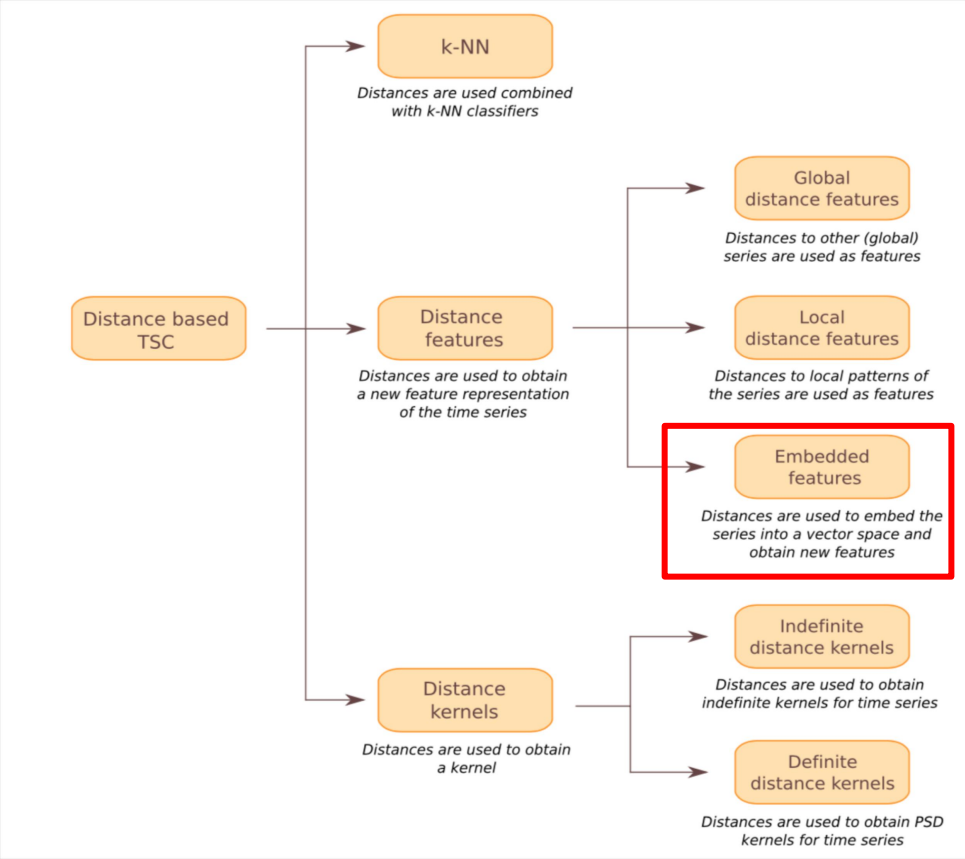


Distance based local features

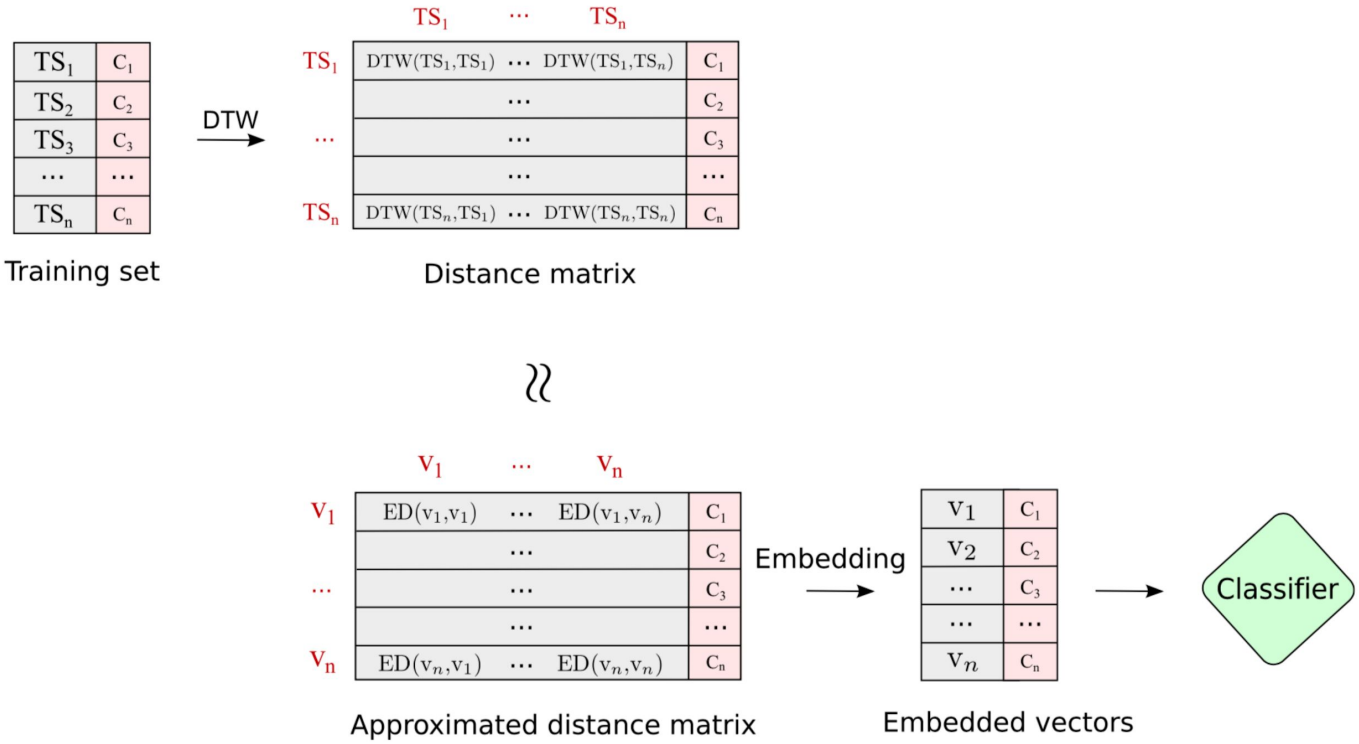


- If shapelets are relevant **very** good results.
- Interpretable.
- Very high computational cost.

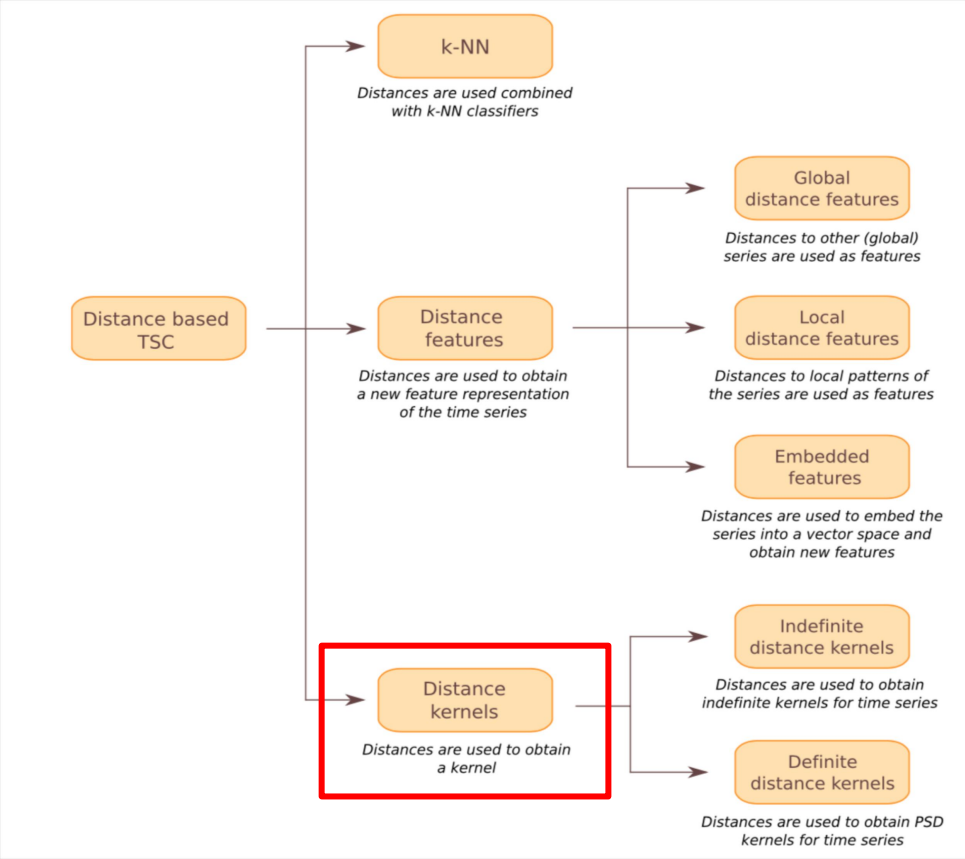
Distance based classification (Abanda el al. 2019)



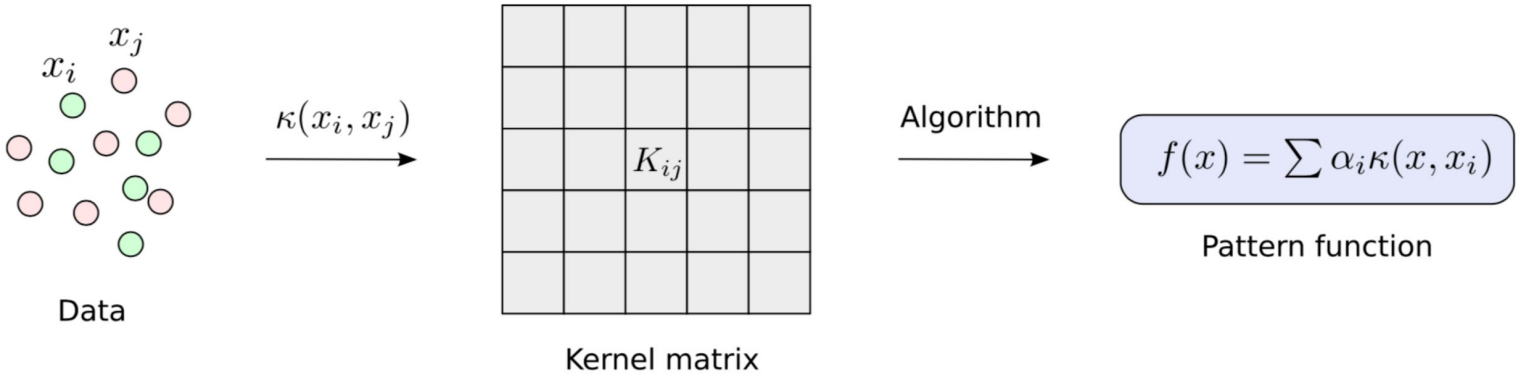
Embedding


TS1 C1
TS2 C2
TS3 C3
... ...
TSn Cn
Training set
DTW
TS1 ... TSn
TS1 DTW(TS1,TS1) ... DTW(TS1,TSn) C1
... C2
... C3
... ...
TSn DTW(TSn,TS1) ... DTW(TSn,TSn) Cn
Distance matrix
≈
V1 ... Vn
V1 ED(v1,v1) ... ED(v1,vn) C1
... C2
... C3
... ...
Vn ED(vn,v1) ... ED(vn,vn) Cn
Approximated distance matrix
Embedding
V1 C1
V2 C2
... C3
... ...
Vn Cn
Embedded vectors
Classifier

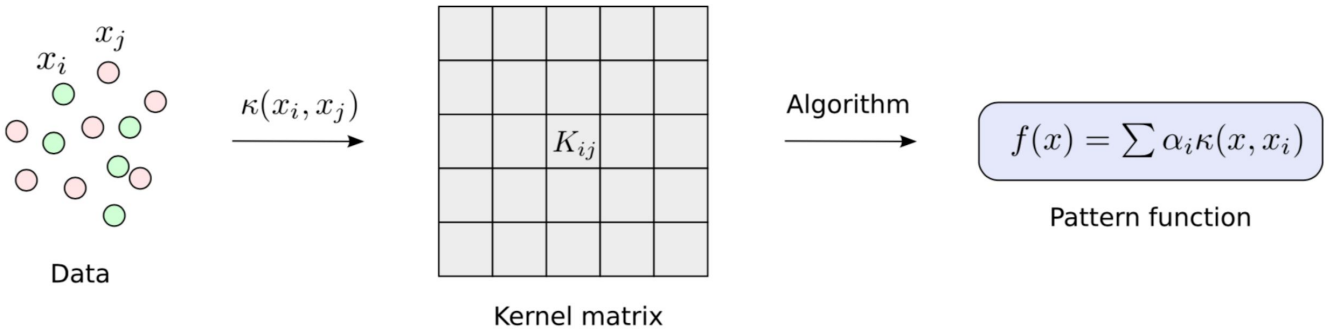
Distance based classification (Abanda el al. 2019)



Kernels


x_i
x_j
Data
$\kappa(x_i, x_j)$
K_{ij}
Kernel matrix
Algorithm
$f(x) = \sum \alpha_i \kappa(x, x_i)$
Pattern function

Kernels



- Algorithms such as SVM or GP can be used.

- Difficulties in obtaining/proving the semi-definiteness of the kernels.
- Theoretical properties are lost with indefinite kernels.

Software

- Distances for time series:

  TSdist
  dtw

- Classification algorithms

  caret

Software

- Distances for time series:

  sklearn.metrics

- Classification algorithms:

  aeon (specific)
  sktime (specific)
  pyts (specific)
  sklearn.classification (non-specific)

Bibliography

- Philippe Esling and Carlos Agon. 2012. Time-series data mining. ACM Comput. Surv. 45, 1, Article 12 (December 2012), 34 pages.

- Bagnall, A., Lines, J., Bostrom, A. et al. The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances. Data Min Knowl Disc 31, 606–660 (2017). https://doi.org/10.1007/s10618-016-0483-9

- Abanda, A., Mori, U. & Lozano, J.A. A review on distance based time series classification. Data Min Knowl Disc 33, 378–412 (2019). https://doi.org/10.1007/s10618-018-0596-4

PROJECT 3:

"You and your team have created a univariate time series classification algorithm and intend to send ***a short paper to a conference****. The aim of the paper is to show that your algorithm outperforms the state-of-the-art and for this, you must design and execute an experimentation that will serve for this purpose and report the results and findings in a suitable manner. The conference you have chosen is one of the most prestigious conferences in the area of time series analysis, so it only has an acceptance rate of %25. Good luck!"*